

# CLUSTERED INFERENCE FOR DYADIC DATA WITH DEPENDENT LATENT POSITIONS

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## Abstract

Inference procedures for dyadic data based on two-way clustering rely on the data being exchangeable and dissociated. In particular, observations must be independent if they have no index in common. In an effort to relax this we consider, instead, data where  $Y_{ij}$  and  $Y_{pq}$  can be dependent for all index pairs, with the dependence vanishing as the distance between the indices grows large. We establish limit theory for the sample mean and propose analytical and bootstrap procedures to perform inference.

**Keywords:** bootstrap, clustering, dependence, dyadic data, inference, serial correlation

## 1 Introduction

Dyadic data  $Y_{ij}$ , for  $1 \leq i < j \leq n$ , arise naturally in the analysis of networks involving  $n$  nodes. A frequent concern that emerges when dealing with such data is how to take into account the potential dependence between observations when performing statistical inference. Independence across all dyads, i.e., all pairs of units, will often feel too strong to impose. An early contribution where this issue was raised is [Fafchamps and Gubert \(2007\)](#).

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They proposed what has become known as a two-way clustering approach to constructing standard errors. The key assumption underlying the appropriateness of such a procedure is that  $Y_{ij}$  and  $Y_{pq}$  are independent unless they have an index in common.

This dependency structure is a natural generalization of the one-way error-component formulation (Moulton 1986, 1990). It holds generally for a dyadic data array provided that it is exchangeable and dissociated. In that case, by well-known results of Aldous (1981), Hoover (1979), and Kallenberg (1989), the data have a nonparametric representation as in

$$Y_{ij} = h(Z_i, Z_j, Q_{ij})$$

for some function  $h$  and (latent) random variables  $\{Z_i\}$  and  $\{Q_{ij}\}$  that are independent and identically distributed across units and dyads, respectively, and independent of each other. This representation is a two-way random-effect model and forms the basis for many popular models of pairwise interaction. Limit theorems for this type of dependence are available (an early contribution is Silverman 1976).<sup>1</sup> A variety of inference procedures have been proposed based on plug-in estimators of the asymptotic variance, the bootstrap, and the jackknife; see Cameron, Gelbach and Miller (2011), Davezies, D’Haultfoeuille and Guyonvarch (2021), MacKinnon, Nielsen and Webb (2021), and Menzel (2021) for many such proposals.<sup>2</sup>

Dissociatedness means that dependence is restricted to be within the same column and row of the data array along which, by exchangeability, it is necessarily constant. This may be too severe in many applications. As a response to this we consider a generalized version of the representation above by allowing for  $\{Z_i\}$  to be a stationary strong-mixing process. To see how this is helpful suppose that the random effect is a moving-average process of order one. Then a given observation will correlate not only with all observations in its own

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<sup>1</sup>Theory for dyadic data of this sort is closely related to U-statistics. The key is to deal with the dependence across the summands induced by the presence of  $\{Z_i\}$ . This is different from work on random fields, such as Bolthausen (1982) and Jenish and Prucha (2009).

<sup>2</sup>These contributions can, in turn, be seen as generalizations of methods designed for one-way clustering, such as Arellano (1987), Liang and Zeger (1986), Cameron, Gelbach and Miller (2008), and Djogbenou, MacKinnon and Nielsen (2019).

row and column, but also with all observations in the rows right above and below, and the column immediately to the left and right. Furthermore, along the row and diagonal, the correlation is different for the observations immediately adjacent and those further away. Other types of processes, such as autoregressive processes or processes with both an autoregressive and a moving-average component allow for richer forms of dependence and are covered by our setup below. Our concern about the limitations of exchangeability and dissociatedness are related to a similar issue in a panel-data context. There, [Thompson \(2011\)](#) made an early case for the importance of allowing for serial dependence in time effects. Different inference procedures for this case have subsequently been proposed by [Chen and Vogelsang \(2024\)](#), [Chiang, Hansen and Sasaki \(2024\)](#), [Hounyo and Lin \(2025a\)](#), and [Hounyo and Lin \(2025c\)](#).<sup>3</sup>

Within our extended setup we begin by establishing the limit distribution of the sample mean. The form of the asymptotic variance suggests the construction of a standard error by first clustering observations by unit and then constructing a HAC-type estimator as in the time-series literature. Such an estimator is straightforward to implement and we give conditions under which it is consistent. We also propose a bootstrap procedure that takes into account all forms of dependence. Our proposal takes the form of a (circular) resampling procedure over the units to construct a new index set to which dyadic observations are then attributed. This procedure can be seen to be an extension of a particular bootstrap for the graphon due to [Green and Shalizi \(2022\)](#). For this procedure we give two results. The first is the consistency of the variance of the bootstrap distribution (conditional on the data) for the long-run variance of the sample mean. The second is the consistency of the bootstrap distribution for the distribution of the sample mean. The latter can be used to perform inference directly, bypassing the construction of a standard error.

The bootstrap procedures are easier to implement than the HAC estimator and are shown to be consistent under a weaker condition on the block size than what is needed on the bandwidth for the HAC estimator. We also provide a data-driven procedure to select

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<sup>3</sup>Other authors have raised related concerns about the strong homogeneity restrictions implicit in the Aldous-Hoover-Kallenberg device. An example is the recent work by [Yap \(2025\)](#).

both of these tuning parameters, borrowed from [Politis and White \(2004\)](#), and test our procedures both with fixed and data-driven tuning parameters in a large set of simulation designs. Based on these, the bootstrap, and particularly the reverse-percentile approach, carries our preference.

An important point of note is that our mixing conditions imply that the nodes exhibit an ordering. Such an ordering is natural in a time-series context, such as sequential entry of individuals into a network or of firms into a market, or in referral networks, for example. It is also appropriate in certain spatial contexts, where units geographically spread out over a line such as a river or coastline. More generally, it arises when nodes can be ranked according to some observable attribute. In contrast, our results here will not be suitable for situations where dependence is governed by some distance measure as in [Conley \(1999\)](#) and [Conley, Gonçalves, Kim and Perron \(2023\)](#), say. We discuss this, as well as other possible extensions and generalizations, in our concluding remarks at the end of the paper. All proofs are collected in the Appendix.

## 2 Limit behavior of the sample mean

For  $1 \leq i \neq j \leq n$  we observe the dyadic variable

$$Y_{ij} = h(Z_i, Z_j, Q_{ij}),$$

where the sequence  $\{Z_i\}$  is stationary and strong mixing, the sequence  $\{Q_{ij}\}$  is independent and identically distributed, and the sequences  $\{Q_{ij}\}$  and  $\{Z_i\}$  are independent of one another. The data are undirected and so  $h$  is symmetric in its first two arguments, but is otherwise unspecified. Because all its arguments are latent it is without any loss of generality to impose that the marginal distribution of  $Z_i$  and of  $Q_{ij}$  is uniform on the interval  $[0, 1]$ .

We are interested in the behavior, in an asymptotic where  $n \rightarrow \infty$ , of the sample mean,

$$\bar{Y}_n = \frac{2}{n(n-1)} \sum_{i=1}^n \sum_{i < j} Y_{ij}.$$

Without loss of generality, we will consider the scalar case throughout. We will derive the large-sample distribution of  $\bar{Y}_n$  under a set of regularity conditions. To state them, introduce

$$h_2(Z_i, Z_j) = \mathbb{E}(Y_{ij}|Z_i, Z_j),$$

and let

$$\alpha_\tau = \sup_{A \in \mathcal{A}_i} \sup_{B \in \mathcal{B}_{i+\tau}} |\mathbb{P}(A \cap B) - \mathbb{P}(A) \mathbb{P}(B)|,$$

for  $\mathcal{A}_i$  and  $\mathcal{B}_i$  the sigma algebras generated by the sequences  $Z_i, Z_{i-1}, \dots$  and  $Z_i, Z_{i+1}, \dots$

**Assumption 1.**

(i) *There is a finite constant  $c > 0$  so that*

$$|h_2(z', z) - h_2(z'', z)| \leq c |z' - z''|$$

*for all  $z, z', z''$ .*

(ii) *There is a  $\delta > 0$  and a finite constant  $c$  so that  $\mathbb{E}(|Y_{ij}|^2) < c$  and*

$$\mathbb{E}(|h_2(Z_i, Z_{i+\tau})|^{2+\delta}) < c$$

*for all  $\tau$ .*

(iii) *The mixing coefficients satisfy  $\alpha_\tau = O(\tau^{-\rho})$  for  $\rho > 3\delta+5/2\delta$  with  $\delta > 0$  as given in (ii).*

Assumption 1 collects a Lipschitz-continuity condition along with conventional moment conditions and an associated rate of decay on the strong-mixing coefficients of the latent process  $\{Z_i\}$ .

The smoothness requirement concerns the kernel  $h_2$ , and not the link function  $h$ . To illustrate the difference, consider the baseline model for non-cooperative network formation

$$Y_{ij} = \{Z_i > Q_{ij}\} \{Z_j > Q_{ij}\},$$

where  $\{\cdot\}$  denotes the indicator function. Then  $h_2(z, z') = \min(z, z')$ , which is Lipschitz continuous. Similarly, typical specifications for cooperative link formation are variations of

$$Y_{ij} = \{|Z_i - Z_j| < Q_{ij}\}.$$

Integrating out  $Q_{ij}$  thus leads to  $h_2(z, z') = 1 - |z - z'|$ , which again satisfies Assumption 1(i). Assumption 1(i) is used to deal with certain remainder terms under strong mixing. Related work (Chiang, Hansen and Sasaki 2024; Hounyo and Lin 2025a) instead strengthens strong mixing to absolutely regularity to do so.

The restriction  $\mathbb{E}(|Y_{ij}|^2) < c$  allows to perform a decomposition along the lines of Hoeffding (1948) and is neither particularly strong nor surprising given our interest in a limit distribution. The moment demand on the kernel is again helpful in dealing with remainder terms. A sufficient condition for Assumption 1(ii) then is that  $\mathbb{E}(|Y_{ij}|^{2+\delta}) < c$ , which is mild.

The requirement on the rate of decay on the strong-mixing coefficients, finally, is stated at a high level, but sufficient conditions are well understood and available for many processes. For example, Assumption 1(iii) is known to be satisfied for any  $\rho > 0$  in autoregressive/moving-average processes if the innovations have an absolutely-continuous distribution (Mokkadem 1988).

The key step in obtaining the asymptotic behavior of the sample mean is showing that

$$\sqrt{n}(\bar{Y}_n - \theta) = \frac{2}{\sqrt{n}} \sum_{i=1}^n X_i + o_P(1), \quad (2.1)$$

where  $\theta = \mathbb{E}(h_1(Z_i))$  and  $X_i = h_1(Z_i) - \theta$  for  $h_1(z) = \mathbb{E}(h_2(z, Z_j))$ . Under Assumption 1 the long-run variance

$$\sigma^2 = \sum_{\tau=-\infty}^{\infty} \omega_{\tau}, \quad \omega_{\tau} = \mathbb{E}(X_i X_{i+\tau}),$$

is well defined. The following theorem, whose proof is given in the Appendix, is then obtained.

**Theorem 1.** *Let Assumption 1 hold and suppose that  $\sigma^2 > 0$ . Then*

$$\sqrt{n}(\bar{Y}_n - \theta) \xrightarrow{L} N(0, 4\sigma^2)$$

as  $n \rightarrow \infty$

Inspection of the proof reveals that Theorem 1 continuous to go through as stated if  $\{Q_{ij}\}$  is weakly dependent across dyads. The only high-level requirement for this to be so is that

$$\frac{2}{n(n-1)} \sum_{i=1}^n \sum_{i < j} (Y_{ij} - h_2(Z_i, Z_j)) = o_P(n^{-1/2}),$$

which is compatible with many forms of dependence. We retain independence here merely to focus on the impact of serial dependence in  $\{Z_i\}$  vis-à-vis the setting of exchangeable and dissociated data.

Having  $\sigma^2 > 0$  rules out the degenerate case, which would yield a different convergence rate and limit distribution. To illustrate, suppose that  $Y_{ij} = (Z_i - 1/2)(Z_j - 1/2) + (Q_{ij} - 1/2)$ , recalling that all latent variables are normalized to be uniform on  $[0, 1]$ . Then  $\mathbb{E}(Z_i) = 1/2$ , and so  $X_i$  has a degenerate distribution at zero. Therefore,  $\sqrt{n}(\bar{Y}_n - \theta) \xrightarrow{P} 0$ . Nevertheless,

$$n\bar{Y}_n = \left( \frac{1}{\sqrt{n}} \sum_{i=1}^n (Z_i - 1/2) \right)^2 - \left( \frac{1}{n} \sum_{i=1}^n (Z_i - 1/2)^2 \right) + \left( \frac{1}{(n-1)} \sum_{i=1}^n \sum_{j \neq i} (Q_{ij} - 1/2) \right) + o_P(1).$$

Under Assumption 1, we have that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (Z_i - 1/2) \xrightarrow{L} N(0, \gamma^2), \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n (Z_i - 1/2)^2 \xrightarrow{P} 1/12,$$

where  $\gamma^2 = \sum_{\tau=-\infty}^{\infty} \mathbb{E}((Z_i - 1/2)(Z_{i+\tau} - 1/2))$  and we have used the fact that the uniform distribution has variance  $1/12$ . In the same way,  $1/(n-1) \sum_{i=1}^n \sum_{j \neq i} (Q_{ij} - 1/2) \xrightarrow{L} N(0, 1/12)$ . Therefore,

$$n\bar{Y}_n \xrightarrow{L} \gamma^2 \chi_1^2 - 1/12 + N(0, 1/12),$$

which is clearly non-normal, in general. This example is an adaptation of a standard example from the U-statistics literature. Indeed, the issue of degeneracy is well-studied in that literature; see, e.g., [van der Vaart \(2000, Section 12.3\)](#) for a textbook treatment and [Leucht \(2012\)](#) for results under dependence. In the current context the situation is more complicated due to the presence of  $\{Q_{ij}\}$ , which is not ignorable. Conversely, normality at the  $n$ -rate would follow if  $\{Z_i\}$  would be completely absent. Inference in that case would be immediate, as the  $Y_{ij}$  would be i.i.d. across dyads. Of course, the issue is to construct procedures that are adaptive, in the sense that they provide correct inference no matter

the situation one is in. unfortunately, this is not possible—see [McCullagh \(2000\)](#), [Owen \(2007\)](#), [Menzel \(2021\)](#), [Juodis \(2025\)](#), and [Hounyo and Lin \(2025b\)](#) for results in related settings—and so, in a first instance, we will content ourselves with making progress on the non-degenerate case in this paper.

### 3 Inference procedures

We now move to performing inference. We consider two approaches. The first approach operationalizes the limit result in [Theorem 1](#) by combining it with an estimator of the asymptotic variance. Here, we propose two such variance estimators, a plug-in estimator and a bootstrap estimator. The second approach uses the bootstrap distribution directly to construct critical values, confidence intervals, or  $p$ -values.

#### 3.1 HAC variance estimator

To estimate  $\sigma^2$  one option is to use a truncation-based estimator in the spirit of [Newey and West \(1987\)](#), i.e.,

$$\hat{\sigma}^2 = \hat{\omega}_0 + 2 \sum_{\tau=1}^{m_{\text{HAC}}-1} (1 - \tau/m_{\text{HAC}}) \hat{\omega}_\tau,$$

where  $m_{\text{HAC}}$  is a chosen bandwidth parameter and  $\hat{\omega}_\tau$  is an estimator of  $\omega_\tau = \mathbb{E}(X_i X_{i+\tau})$ .

We will use

$$\hat{\omega}_\tau = \frac{1}{n} \sum_{i=1}^{n-\tau} (\tilde{Y}_i - \bar{Y}_n)(\tilde{Y}_{i+\tau} - \bar{Y}_n),$$

which amounts to using the centered sample mean  $(\tilde{Y}_i - \bar{Y}_n) = 1/(n-1) \sum_{j \neq i} (Y_{ij} - \bar{Y}_n)$  as an estimator of  $X_i$ .

To explain the form of  $\hat{\sigma}^2$  note that, when  $m_{\text{HAC}} = 1$  we have  $\hat{\sigma}^2 = \hat{\omega}_0$ , which would be a valid variance estimator when  $\{Z_i\}$  is a sequence of independent variables. In that case it suffices to account for the within-row and within-column dependence in the data, and this is done by the averaging performed when constructing  $(\tilde{Y}_i - \bar{Y}_n)$ , as is conventional in



problems involving U-statistics.<sup>4</sup>

In the present case, we have additional dependence stemming from the serial dependence in the  $Z_i$ . This is handled by adding (kernel-weighted) covariances for all leads and lags ranging from 1 to  $m_{\text{HAC}} - 1$ . As usual, the Bartlett weights are not the only possibility but they ensure that  $\hat{\sigma}^2$  is always non-negative.

The next theorem states a consistency result for this variance estimator.

**Theorem 2.** *Let Assumption 1 hold and suppose that  $m_{\text{HAC}} \rightarrow \infty$  so that  $n^{-1}m_{\text{HAC}}^2 \rightarrow 0$ . Then*

$$\hat{\sigma}^2 \xrightarrow{P} \sigma^2$$

as  $n \rightarrow \infty$ .

For a data-driven selection of the bandwidth parameter in the construction of this estimator we can follow procedures from the time-series literature, such as those proposed by Andrews (1991), Andrews and Monahan (1992), and Newey and West (1994). We discuss this in more detail below.

## 3.2 Bootstrap

The existing cluster-bootstrap procedures are not designed for the mixing structure we entertain here. We take inspiration from Green and Shalizi (2022) and devise a circular version of the moving block bootstrap (Politis and Romano 1992) for dyadic data with dependent node labels.<sup>5</sup>

To state the bootstrap scheme we introduce the notation

$$i \bmod n = i - n \lfloor (i-1)/n \rfloor.$$

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<sup>4</sup>Several cluster-robust variance estimators available in the literature include additional terms in their construction (see, e.g., Graham 2020, Section 4.4). These terms are asymptotically negligible. They may be either positive or negative, their inclusion is not guaranteed to improve finite-sample performance, and may lead to negative variance estimators.

<sup>5</sup>An alternative would be the dependent wild bootstrap of Shao (2010) applied to  $\tilde{Y}_i - \bar{Y}_n$  which, as in our feasible HAC construction, serves as an estimator of  $X_i$ .

For a chosen integer  $b$ , let  $\varpi_1, \dots, \varpi_b$  be independent draws from the uniform distribution on the index set  $\{1, 2, \dots, n\}$ . For each  $1 \leq u \leq b$ , we then construct the block of  $m_{\text{CB}}$  consecutive indices starting at  $\varpi_u$ ,

$$\{\varpi_u, (\varpi_u + 1) \bmod n, \dots, (\varpi_u + m_{\text{CB}} - 1) \bmod n\},$$

and concatenate the  $b$  blocks to form a reshuffled index set. We will maintain for notational simplicity that  $bm_{\text{CB}} = n$ . The approach is, therefore, to randomly draw  $b$  blocks of length  $m_{\text{CB}}$  from the index set  $\{1, 2, \dots, n\}$ , wrapping around in a circle.<sup>6</sup> Each  $1 \leq i \leq n$  in the bootstrapped index set maps to the original index set via

$$\phi_i = (\varpi_{\lceil i/m_{\text{CB}} \rceil} + i - (\lceil i/m_{\text{CB}} \rceil - 1)m_{\text{CB}} - 2) \bmod n + 1.$$

We then generate a bootstrap sample as

$$Y_{ij}^* = \begin{cases} Y_{\phi_i \phi_j} & \text{if } \phi_i \neq \phi_j \\ 0 & \text{if } \phi_i = \phi_j \end{cases}.$$

from which the bootstrap sample mean,

$$\bar{Y}_n^* = \frac{2}{n(n-1)} \sum_{i=1}^n \sum_{i < j} Y_{ij}^*,$$

may then be constructed.

Setting  $Y_{ij}^* = 0$  when  $\phi_i = \phi_j$  is done because there is no corresponding observation in the original data. In the Appendix, Theorems 3 and 4 below are shown to hold when, instead of setting the bootstrap observation to zero, one sets it to a random variable  $D_i$  for which  $\mathbb{E}(D_i^2) < \infty$ . The use of zero effectively means that the observation does not contribute to  $\sum_{i=1}^n \sum_{i < j} Y_{\phi_i \phi_j}$ , and so seems a natural choice. We show in the Appendix that

$$\mathbb{E}^*(\bar{Y}_n^*) = \left( \frac{n - m_{\text{CB}}}{n} \right) \bar{Y}_n + O_P \left( \frac{m_{\text{CB}} - 1}{n} \right) \quad (3.2)$$

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<sup>6</sup>Without wrapping around the unit circle we would be constructing a version of the moving-block bootstrap (Künsch 1989). Under our conditions this is equally valid. However, we follow the literature and consider the circular scheme to alleviate bias in the moving-block bootstrap that arises from the relative undersampling of observations at the beginning and end of the sequence  $\{1, 2, \dots, n\}$ ; see Politis and Romano (1992, 1994).

Here, the first right-hand side term features a shrinkage factor that is due to setting the artificial observations equal to zero. The remainder term, whose precise form is given in the Appendix, appears only when  $m_{\text{CB}} > 1$  and does not depend on how what value we choose for the artificial observations. Given this expression it would be possible, if desired, to adjust  $\bar{Y}_n^*$  to make it (conditionally) unbiased for  $\bar{Y}_n$ . For the case where  $m_{\text{CB}} = 1$  this amounts to a simple rescaling by  $n/(n-1)$ .

The intuition for why our bootstrap scheme works is that, as we show in the Appendix,

$$\sqrt{n}(\bar{Y}_n^* - \bar{Y}_n) = \frac{2}{\sqrt{n}} \sum_{i=1}^n X_i^* + o_{P^*}(1),$$

where  $X_i^* = X_{\phi_i}$  and  $A_n^* = o_{P^*}(1)$  means that  $\mathbb{P}(\mathbb{P}^*(|A_n^*| > \epsilon^*) > \epsilon) = o(1)$  for all  $\epsilon^* > 0$  and  $\epsilon > 0$ . The dominant term on the right-hand side is a (scaled) sample mean of a stationary mixing process, for which the validity of the traditional circular block bootstrap is well established.

A bootstrap estimator of  $4\sigma^2$  is

$$n \text{var}^*(\bar{Y}_n^*) = n \mathbb{E}^*((\bar{Y}_n^* - \mathbb{E}^*(\bar{Y}_n^*))^2),$$

that is, the variance of the bootstrap distribution, scaled by  $n$ . This estimator, too, is consistent.

**Theorem 3.** *Let Assumption 1 hold and suppose that  $m_{\text{CB}} \rightarrow \infty$  so that  $n^{-1}m_{\text{CB}} \rightarrow 0$ . Then*

$$n \text{var}^*(\bar{Y}_n^*) \xrightarrow{P} 4\sigma^2$$

as  $n \rightarrow \infty$ .

For the choice of  $m_{\text{CB}}$  we can again take inspiration from the existing time-series literature ([Politis and White 2004](#)); see below.

Theorems 3 is qualitatively similar to Theorem 2. In conjunction with Theorem 1, both can be used to construct test statistics that yield asymptotically size-correct inference. One difference of note is that the bootstrap avoids the need to estimate the projection

term  $X_i$ . This explains the weaker rate requirement on  $m_{\text{CB}}$  in Theorem 3 relative to  $m_{\text{HAC}}$  in Theorem 2.

Moving on, the next theorem shows that the bootstrap also consistently estimates the limit distribution of the sample mean. This again is true under the weaker rate requirement that only  $n^{-1}m_{\text{CB}} \rightarrow 0$  and by-passes use of the normal approximation entirely when performing inference.

**Theorem 4.** *Let Assumption 1 hold and suppose that  $m_{\text{CB}} \rightarrow \infty$  so that  $n^{-1}m_{\text{CB}} \rightarrow 0$ . Then*

$$\sup_a |\mathbb{P}^*(\sqrt{n}(\bar{Y}_n^* - \bar{Y}_n) \leq a) - \mathbb{P}(\sqrt{n}(\bar{Y}_n - \theta) \leq a)| \xrightarrow{P} 0$$

as  $n \rightarrow \infty$ .<sup>7</sup>

By van der Vaart (2000, Lemma 23.3), Theorem 4 implies that inference based on the reverse-percentile bootstrap is asymptotically justified. For any  $\alpha \in (0, 1)$ , denote by  $\hat{Q}_\alpha$  the  $\alpha$ -quantile of the distribution of  $\sqrt{n}(\bar{Y}_n^* - \bar{Y}_n)$ , conditional on the original data. Then,

$$\mathbb{P}(\sqrt{n}(\bar{Y}_n - \theta) \leq \hat{Q}_\alpha) \xrightarrow{n \rightarrow \infty} \alpha.$$

As an example, a two-sided confidence set for  $\theta$  with nominal coverage  $1 - \alpha$  is the interval

$$\left\{ \vartheta : \bar{Y}_n - \frac{\hat{Q}_{1-\alpha/2}}{\sqrt{n}} \leq \vartheta \leq \bar{Y}_n - \frac{\hat{Q}_{\alpha/2}}{\sqrt{n}} \right\}$$

for any  $\alpha < 1/2$ . Critical values and  $p$ -values for specific null hypotheses follow from the same argument.

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<sup>7</sup>In light of (3.2) we equally have that

$$\sup_a |\mathbb{P}^*(\sqrt{n}(\bar{Y}_n^* - \mathbb{E}^*(\bar{Y}_n^*)) \leq a) - \mathbb{P}(\sqrt{n}(\bar{Y}_n - \theta) \leq a)| \xrightarrow{P} 0$$

as  $n \rightarrow \infty$ . Such centering is entertained, for example, in Lahiri (1991).

## 4 Numerical illustration

A simulation experiment was conducted to evaluate the performance of the various inference techniques proposed above. Here we report results for data generating processes in which

$$X_i = \varrho X_{i-1} + V_i + \varphi V_{i-1},$$

where  $V_i$  are standardized  $\chi_\eta^2$  random variables. In this way,  $\mathbb{E}(V_i) = 0$  and  $\mathbb{E}(V_i^2) = 1$ , but the distribution of  $V_i$  has skewness  $\sqrt{8/\eta}$ . We consider  $\eta \in \{1, 2, 4\}$ , leading to skewness being 4, 2, and 1, respectively. The tables below give results for different configurations of this ARMA process. The baseline case has no dependence, that is  $(\varrho, \varphi) = (0, 0)$ . We then consider the MA(1) specification  $(\varrho, \varphi) = (0, 1)$  and ARMA(1,1) specifications with varying degree of persistence,  $(\varrho, \varphi) \in \{(0.25, 1), (0.50, 1), (0.75, 1)\}$ . To conserve on space we fix the sample size to 250 throughout.

Tables 1–3 contain simulations results based on 5,000 Monte Carlo replications and 599 bootstrap replications. For each design the tables provide the (Monte Carlo) standard deviation of the sample mean along with the (average) estimated standard error obtained through the plug-in estimator (HAC) and the bootstrap (CB). Next, the coverage rate and average length of three confidence intervals (with theoretical coverage of .95) are given. The first two (HAC and CB) are based on the normal approximation together with the respective variance estimator. The third, finally, (CB%) is based on the reverse-percentile bootstrap as described above. For each design we report on results obtained through different bandwidth and block-size choices. We describe this part of the implementation next.

The data-driven proposals of Newey and West (1994) and Politis and White (2004) to selecting the bandwidth  $m_{\text{HAC}}$  and block size  $m_{\text{CB}}$ , respectively, are both of the same form

$$\lfloor (3/2)^{1/3} (|\hat{s}_1/\hat{s}_0|)^{2/3} n^{1/3} \rfloor,$$

where  $\hat{s}_1$  and  $\hat{s}_0$  are estimators of  $s_1 = 2 \sum_{\tau=1}^{\infty} \tau \omega_\tau$  and  $s_0 = \omega_0 + 2 \sum_{\tau=1}^{\infty} \omega_\tau = \sigma^2$ . From calculations in Hannan (1970, pp. 286) and Priestley (1981, pp. 568) this is an estimator of the mean-squared-error minimising choice. The difference between Newey and West (1994)

and [Politis and White \(2004\)](#) lies in the construction of  $\hat{s}_1$  and  $\hat{s}_0$ . In numerical experiments (not reported here) we found that the former's suggestion tends to be too conservative in our setting, and so we focus on the latter. [Politis and White \(2004\)](#) suggest the estimators

$$\hat{s}_1 = 2 \sum_{\tau=1}^b \tau \varpi(\tau/b) \hat{\omega}_\tau, \quad \hat{s}_0 = \hat{\omega}_0 + 2 \sum_{\tau=1}^b \varpi(\tau/b) \hat{\omega}_\tau,$$

with  $\varpi(t) = 1 \{0 \leq |t| \leq 1/2\} + 2(1 - |t|) \{1/2 \leq |t| \leq 1\}$  the 'flat-top' kernel. In this expression, the only remaining parameter is  $b = 2q$ , where  $q$  is the smallest integer for which

$$|\hat{\omega}_{q+\tau}| < 2 \hat{\omega}_0 \sqrt{\log n/n},$$

with  $\tau$  ranging from 1 to  $\max\{5, \lfloor \sqrt{\log n} \rfloor\}$ . This approach gives the same choice for both the plug-in variance estimator and the bootstrap; we will refer to it simply as  $\hat{m}$  in our discussion to follow. To evaluate the adequacy of  $\hat{m}$  we also provide results for four fixed choices; they are 1, 5, 10, and 20. The first of these amounts to not accounting for serial correlation at all. The three other values were taken to provide a range. In the tables below, for each  $(\varrho, \varphi)$  design, the last row contains the results based on  $\hat{m}$ ; the entry for the column called  $m$  gives the median value for the bandwidth/block length that was selected; for example,  $\hat{1}$  means that the median bandwidth/block length in this particular design was equal to one.

Inspection of the tables reveals that the results are quite stable across the different skewness levels considered. When  $(\varrho, \varphi) = (0, 0)$  the data are exchangeable and dissociated;  $m = 1$  is the correct choice here and the procedure of [Politis and White \(2004\)](#) picks this up well. Nonetheless, the actual variance estimators and coverage intervals all perform well for all values of  $m$  in the table in this case.

In the MA(1) model dependence is present but limited. Mistakingly treating the data as dissociated- by setting  $m$  equal to one leads to underestimation of the actual variability and a coverage rate that is too low. Working with any  $m > 1$  rectifies both problems here, and  $\hat{m}$  tends to be around 5 over the simulations.

The underestimation problem is more severe when we add an autoregressive component and, as expected, the distortion increases as  $\varrho$  increases. Here, an increasingly larger  $m$  is

Table 1: Simulation results for  $\eta = 1$  (skewness of 4)

| DESIGN    |           |            | STD ERROR |       |       | CI COVERAGE (95%) |       |       | CI LENGTH |       |       |
|-----------|-----------|------------|-----------|-------|-------|-------------------|-------|-------|-----------|-------|-------|
| $\varrho$ | $\varphi$ | $m$        | MC        | HAC   | CB    | HAC               | CB    | CB%   | HAC       | CB    | CB%   |
| .00       | 0         | 1          | 0.127     | 0.125 | 0.125 | 0.945             | 0.942 | 0.939 | 0.491     | 0.489 | 0.488 |
| .00       | 0         | 5          | 0.127     | 0.124 | 0.124 | 0.938             | 0.938 | 0.934 | 0.486     | 0.485 | 0.483 |
| .00       | 0         | 10         | 0.127     | 0.122 | 0.122 | 0.934             | 0.934 | 0.932 | 0.479     | 0.477 | 0.476 |
| .00       | 0         | 20         | 0.127     | 0.119 | 0.126 | 0.923             | 0.938 | 0.948 | 0.466     | 0.493 | 0.492 |
| .00       | 0         | $\hat{1}$  | 0.127     | 0.125 | 0.125 | 0.942             | 0.941 | 0.938 | 0.490     | 0.489 | 0.488 |
| .00       | 1         | 1          | 0.254     | 0.177 | 0.176 | 0.819             | 0.818 | 0.817 | 0.692     | 0.690 | 0.689 |
| .00       | 1         | 5          | 0.254     | 0.234 | 0.233 | 0.925             | 0.925 | 0.921 | 0.919     | 0.915 | 0.913 |
| .00       | 1         | 10         | 0.254     | 0.237 | 0.236 | 0.925             | 0.923 | 0.922 | 0.931     | 0.925 | 0.923 |
| .00       | 1         | 20         | 0.254     | 0.234 | 0.247 | 0.918             | 0.935 | 0.945 | 0.917     | 0.969 | 0.966 |
| .00       | 1         | $\hat{5}$  | 0.254     | 0.235 | 0.235 | 0.925             | 0.926 | 0.923 | 0.921     | 0.920 | 0.918 |
| .25       | 1         | 1          | 0.339     | 0.204 | 0.203 | 0.755             | 0.753 | 0.746 | 0.798     | 0.795 | 0.794 |
| .25       | 1         | 5          | 0.339     | 0.300 | 0.299 | 0.914             | 0.911 | 0.907 | 1.177     | 1.172 | 1.170 |
| .25       | 1         | 10         | 0.339     | 0.311 | 0.308 | 0.919             | 0.918 | 0.916 | 1.217     | 1.209 | 1.206 |
| .25       | 1         | 20         | 0.339     | 0.309 | 0.326 | 0.916             | 0.931 | 0.944 | 1.210     | 1.278 | 1.273 |
| .25       | 1         | $\hat{5}$  | 0.339     | 0.304 | 0.304 | 0.916             | 0.913 | 0.912 | 1.193     | 1.191 | 1.188 |
| .50       | 1         | 1          | 0.508     | 0.248 | 0.248 | 0.657             | 0.655 | 0.656 | 0.974     | 0.970 | 0.969 |
| .50       | 1         | 5          | 0.508     | 0.414 | 0.412 | 0.884             | 0.883 | 0.879 | 1.624     | 1.615 | 1.613 |
| .50       | 1         | 10         | 0.508     | 0.447 | 0.444 | 0.910             | 0.906 | 0.902 | 1.754     | 1.740 | 1.735 |
| .50       | 1         | 20         | 0.508     | 0.454 | 0.478 | 0.907             | 0.924 | 0.940 | 1.778     | 1.876 | 1.869 |
| .50       | 1         | $\hat{7}$  | 0.508     | 0.445 | 0.446 | 0.904             | 0.905 | 0.907 | 1.746     | 1.749 | 1.745 |
| .75       | 1         | 1          | 1.013     | 0.348 | 0.346 | 0.496             | 0.495 | 0.493 | 1.362     | 1.357 | 1.355 |
| .75       | 1         | 5          | 1.013     | 0.665 | 0.661 | 0.792             | 0.788 | 0.787 | 2.608     | 2.591 | 2.587 |
| .75       | 1         | 10         | 1.013     | 0.788 | 0.779 | 0.861             | 0.857 | 0.850 | 3.087     | 3.055 | 3.048 |
| .75       | 1         | 20         | 1.013     | 0.851 | 0.894 | 0.883             | 0.900 | 0.920 | 3.335     | 3.506 | 3.494 |
| .75       | 1         | $\hat{13}$ | 1.013     | 0.839 | 0.837 | 0.877             | 0.880 | 0.887 | 3.289     | 3.279 | 3.270 |

Results are based on 5000 Monte Carlo replications and 599 bootstrap replications. Last row for each design is based on data-driven bandwidth/block size selection,  $\hat{m}$ . Sample size is 250.

Table 2: Simulation results for  $\eta = 4$  (skewness of 2)

| DESIGN    |           |            | STD ERROR |       |       | CI COVERAGE (95%) |       |       | CI LENGTH |       |       |
|-----------|-----------|------------|-----------|-------|-------|-------------------|-------|-------|-----------|-------|-------|
| $\varrho$ | $\varphi$ | $m$        | MC        | HAC   | CB    | HAC               | CB    | CB%   | HAC       | CB    | CB%   |
| .00       | 0         | 1          | 0.125     | 0.126 | 0.125 | 0.952             | 0.951 | 0.948 | 0.493     | 0.491 | 0.491 |
| .00       | 0         | 5          | 0.125     | 0.124 | 0.124 | 0.947             | 0.945 | 0.944 | 0.487     | 0.485 | 0.485 |
| .00       | 0         | 10         | 0.125     | 0.122 | 0.122 | 0.940             | 0.938 | 0.935 | 0.480     | 0.477 | 0.477 |
| .00       | 0         | 20         | 0.125     | 0.119 | 0.126 | 0.923             | 0.938 | 0.954 | 0.466     | 0.492 | 0.491 |
| .00       | 0         | $\hat{1}$  | 0.125     | 0.125 | 0.125 | 0.949             | 0.948 | 0.947 | 0.490     | 0.489 | 0.489 |
| .00       | 1         | 1          | 0.250     | 0.177 | 0.176 | 0.832             | 0.829 | 0.826 | 0.694     | 0.692 | 0.691 |
| .00       | 1         | 5          | 0.250     | 0.235 | 0.234 | 0.930             | 0.928 | 0.928 | 0.920     | 0.916 | 0.915 |
| .00       | 1         | 10         | 0.250     | 0.238 | 0.236 | 0.931             | 0.926 | 0.925 | 0.932     | 0.925 | 0.924 |
| .00       | 1         | 20         | 0.250     | 0.234 | 0.247 | 0.918             | 0.933 | 0.950 | 0.917     | 0.967 | 0.965 |
| .00       | 1         | $\hat{5}$  | 0.250     | 0.235 | 0.234 | 0.930             | 0.929 | 0.932 | 0.920     | 0.919 | 0.918 |
| .25       | 1         | 1          | 0.333     | 0.204 | 0.203 | 0.768             | 0.766 | 0.761 | 0.799     | 0.797 | 0.797 |
| .25       | 1         | 5          | 0.333     | 0.301 | 0.299 | 0.918             | 0.916 | 0.913 | 1.179     | 1.173 | 1.172 |
| .25       | 1         | 10         | 0.333     | 0.311 | 0.308 | 0.925             | 0.923 | 0.921 | 1.219     | 1.209 | 1.208 |
| .25       | 1         | 20         | 0.333     | 0.309 | 0.325 | 0.916             | 0.929 | 0.949 | 1.210     | 1.275 | 1.273 |
| .25       | 1         | $\hat{5}$  | 0.333     | 0.304 | 0.303 | 0.920             | 0.918 | 0.916 | 1.191     | 1.189 | 1.188 |
| .50       | 1         | 1          | 0.499     | 0.249 | 0.248 | 0.669             | 0.667 | 0.664 | 0.975     | 0.972 | 0.972 |
| .50       | 1         | 5          | 0.499     | 0.415 | 0.413 | 0.889             | 0.888 | 0.886 | 1.627     | 1.617 | 1.616 |
| .50       | 1         | 10         | 0.499     | 0.448 | 0.444 | 0.912             | 0.909 | 0.909 | 1.756     | 1.740 | 1.737 |
| .50       | 1         | 20         | 0.499     | 0.454 | 0.477 | 0.911             | 0.923 | 0.943 | 1.779     | 1.872 | 1.869 |
| .50       | 1         | $\hat{7}$  | 0.499     | 0.444 | 0.446 | 0.909             | 0.909 | 0.915 | 1.742     | 1.747 | 1.744 |
| .75       | 1         | 1          | 0.993     | 0.348 | 0.347 | 0.504             | 0.503 | 0.502 | 1.364     | 1.360 | 1.359 |
| .75       | 1         | 5          | 0.993     | 0.666 | 0.662 | 0.806             | 0.803 | 0.796 | 2.611     | 2.594 | 2.592 |
| .75       | 1         | 10         | 0.993     | 0.788 | 0.779 | 0.867             | 0.861 | 0.861 | 3.090     | 3.054 | 3.049 |
| .75       | 1         | 20         | 0.993     | 0.851 | 0.892 | 0.890             | 0.904 | 0.925 | 3.335     | 3.498 | 3.490 |
| .75       | 1         | $\hat{13}$ | 0.993     | 0.838 | 0.837 | 0.882             | 0.884 | 0.895 | 3.286     | 3.279 | 3.272 |

Results are based on 5000 Monte Carlo replications and 599 bootstrap replications. Last row for each design is based on data-driven bandwidth/block size selection,  $\hat{m}$ . Sample size is 250.



Table 3: Simulation results for  $\eta = 8$  (skewness of 1)

| DESIGN    |           |            | STD ERROR |       |       | CI COVERAGE (95%) |       |       | CI LENGTH |       |       |
|-----------|-----------|------------|-----------|-------|-------|-------------------|-------|-------|-----------|-------|-------|
| $\varrho$ | $\varphi$ | $m$        | MC        | HAC   | CB    | HAC               | CB    | CB%   | HAC       | CB    | CB%   |
| .00       | 0         | 1          | 0.124     | 0.126 | 0.125 | 0.954             | 0.951 | 0.949 | 0.493     | 0.491 | 0.491 |
| .00       | 0         | 5          | 0.124     | 0.124 | 0.124 | 0.949             | 0.947 | 0.947 | 0.486     | 0.484 | 0.484 |
| .00       | 0         | 10         | 0.124     | 0.122 | 0.122 | 0.943             | 0.942 | 0.941 | 0.479     | 0.477 | 0.476 |
| .00       | 0         | 20         | 0.124     | 0.119 | 0.126 | 0.929             | 0.940 | 0.956 | 0.465     | 0.492 | 0.492 |
| .00       | 0         | $\hat{1}$  | 0.124     | 0.125 | 0.125 | 0.952             | 0.952 | 0.951 | 0.490     | 0.489 | 0.489 |
| .00       | 1         | 1          | 0.247     | 0.177 | 0.176 | 0.836             | 0.832 | 0.831 | 0.694     | 0.692 | 0.691 |
| .00       | 1         | 5          | 0.247     | 0.234 | 0.233 | 0.934             | 0.935 | 0.934 | 0.919     | 0.914 | 0.914 |
| .00       | 1         | 10         | 0.247     | 0.237 | 0.236 | 0.936             | 0.933 | 0.935 | 0.930     | 0.924 | 0.923 |
| .00       | 1         | 20         | 0.247     | 0.233 | 0.247 | 0.927             | 0.937 | 0.952 | 0.915     | 0.966 | 0.965 |
| .00       | 1         | $\hat{5}$  | 0.247     | 0.234 | 0.234 | 0.934             | 0.935 | 0.936 | 0.918     | 0.917 | 0.916 |
| .25       | 1         | 1          | 0.328     | 0.204 | 0.203 | 0.767             | 0.767 | 0.765 | 0.799     | 0.797 | 0.797 |
| .25       | 1         | 5          | 0.328     | 0.300 | 0.299 | 0.923             | 0.923 | 0.920 | 1.178     | 1.171 | 1.170 |
| .25       | 1         | 10         | 0.328     | 0.310 | 0.308 | 0.930             | 0.928 | 0.928 | 1.217     | 1.208 | 1.206 |
| .25       | 1         | 20         | 0.328     | 0.308 | 0.325 | 0.923             | 0.935 | 0.951 | 1.207     | 1.275 | 1.273 |
| .25       | 1         | $\hat{5}$  | 0.328     | 0.303 | 0.303 | 0.924             | 0.924 | 0.923 | 1.189     | 1.186 | 1.185 |
| .50       | 1         | 1          | 0.492     | 0.249 | 0.248 | 0.673             | 0.673 | 0.670 | 0.974     | 0.972 | 0.972 |
| .50       | 1         | 5          | 0.492     | 0.414 | 0.412 | 0.895             | 0.892 | 0.892 | 1.624     | 1.614 | 1.614 |
| .50       | 1         | 10         | 0.492     | 0.447 | 0.443 | 0.920             | 0.915 | 0.915 | 1.752     | 1.738 | 1.735 |
| .50       | 1         | 20         | 0.492     | 0.453 | 0.477 | 0.915             | 0.930 | 0.947 | 1.775     | 1.871 | 1.868 |
| .50       | 1         | $\hat{7}$  | 0.492     | 0.444 | 0.445 | 0.915             | 0.915 | 0.921 | 1.739     | 1.744 | 1.742 |
| .75       | 1         | 1          | 0.980     | 0.348 | 0.346 | 0.513             | 0.513 | 0.509 | 1.362     | 1.358 | 1.357 |
| .75       | 1         | 5          | 0.980     | 0.665 | 0.660 | 0.808             | 0.804 | 0.806 | 2.606     | 2.588 | 2.586 |
| .75       | 1         | 10         | 0.980     | 0.786 | 0.778 | 0.874             | 0.870 | 0.871 | 3.083     | 3.050 | 3.044 |
| .75       | 1         | 20         | 0.980     | 0.849 | 0.892 | 0.892             | 0.911 | 0.929 | 3.326     | 3.495 | 3.488 |
| .75       | 1         | $\hat{13}$ | 0.980     | 0.837 | 0.835 | 0.888             | 0.893 | 0.898 | 3.281     | 3.272 | 3.265 |

Results are based on 5000 Monte Carlo replications and 599 bootstrap replications. Last row for each design is based on data-driven bandwidth/block size selection,  $\hat{m}$ . Sample size is 250.

needed to get standard errors that are close to the actual standard deviation of the sample mean. Nevertheless, except perhaps in the most challenging design, inference is close to size correct for all  $m > 1$ .

The plug-in variance estimator and bootstrap variance estimator both perform well. An interesting observation that arises from the tables is that the bootstrap tends to perform noticeably better at the larger values for  $m$ . For the smaller bandwidths, and as well as for the data-driven choice, they behave very similarly; this is in line with their asymptotic equivalence in a related setting (Politis and Romano 1994; Gonçalves and White 2002). The confidence intervals based directly on the bootstrap percentiles, in turn, are always at least as good as those based on the normal approximation, and they tend to perform better when the block length is larger; this is particularly helpful in the more persistent designs.

The data-driven selection of  $\hat{m}$  adapts sensibly to the dependence structure, selecting larger values as the persistence in the data increases. The resulting coverage and interval lengths are typically close to those obtained using the best-performing fixed choice in each of the designs.

To summarise, the simulations, therefore, point in favour of the bootstrap to perform inference. Moreover, the bootstrap variance estimator tends to have a smaller bias than the plug-in estimator, whereas confidence intervals based directly on the bootstrap distribution tend to perform best while, at the same time, being the least sensitive to the choice of block length.

## 5 Concluding remarks

In this paper we have looked at the problem of inference with dyadic data. The conventional setting implies that dependence between  $Y_{ij}$  and  $Y_{pq}$  is present only when  $i < j$  and  $p < q$  have an index in common, and that their covariance in such a case is independent of the indices. Such a setup can be too restrictive in a variety of practical applications. Our framework allows for dependence between  $Y_{ij}$  and  $Y_{pq}$  for all values of the indices. The dependency structure between non-overlapping index pairs is ergodic, and we stated

conditions under which this ensures the sample mean to converge to a Gaussian process at the usual  $n^{-1/2}$  convergence rate. To perform inference, we gave a plug-in variance estimator that generalizes the usual two-way clustering approach and devised a new bootstrap scheme that can be used to estimate both the asymptotic variance and the quantiles of the limit distribution.

Extensions in various directions are possible. First, a directed version of our setup can be set up by having a pair of variables  $Z_i = (Z_i^o, Z_i^{oo})$ , and then asymmetric dyadic variables

$$Y_{ij} = h(Z_i^o, Z_j^{oo}, Q_{ij}).$$

Here, subject to suitable regularity conditions we will have a representation for the sample mean as

$$\sqrt{n}(\bar{Y}_n - \theta) + \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i^o + X_i^{oo}) + o_P(1),$$

where, now,  $X_i^o = h_1^o(Z_i) - \theta$  and  $X_i^{oo} = h_1^{oo}(Z_i) - \theta$  for functions  $h_1^o(z) = \mathbb{E}(h(z, Z_j, Q_{ij}))$  and  $h_1^{oo}(z) = \mathbb{E}(h(Z_i, z, Q_{ij}))$ . The asymptotic variance of the sample mean will then be equal to the long-run variance of  $X_i^o + X_i^{oo}$ , for which a HAC or bootstrap estimator can again be constructed.

Second, there is no apparent reason why our results could not be extended to multi-adic data, where we observe  $Y_{\mathbf{i}}$  for tuples  $\mathbf{i} = (i_1, i_2, \dots, i_d)$  of size  $d > 2$ . Such data are studied in [Owen and Eckles \(2012\)](#) and [Davezies, D'Haultfœuille and Guyonvarch \(2021\)](#), where they are assumed to be exchangeable and dissociated. To build in additional dependence, the same intuition as here applies. In particular, subject to regularity conditions, we should have

$$\sqrt{n}(\bar{Y}_n - \theta) + \frac{d}{\sqrt{n}} \sum_{i=1}^n X_i + o_P(1),$$

where, now, the sample mean is computed over all unique  $d$ -tuples  $\mathbf{i}$ ,  $X_i = h_1(Z_i) - \theta$  as before with  $h_1$  now corresponding to the function obtained on integrating out all random variables but  $Z_i$  from the [Kallenberg \(1989\)](#) representation of the data array; see, e.g., Equation (2.1) in [Davezies, D'Haultfœuille and Guyonvarch \(2021\)](#) for this representation.

Third, it would be valuable to accommodate other forms of dependence that do not rely on an ordering of the observations. Forms of such dependence are entertained (in a

different context) in [Conley \(1999\)](#) and [Conley, Gonçalves, Kim and Perron \(2023\)](#). To deal with this type of a dependence here, one would need to establish a version of (2.1) and, in particular, the negligibility of the remainder term. One would then be able to construct a robust variance estimator using their proposed approaches, only here again based on an estimator of the summand in the leading term of this expansion, as is done here in the HAC context.

## Appendix

### Proof of Theorem 1

We begin by defining

$$V_{ij} = Y_{ij} - \theta - X_i - X_j$$

where, recall,  $X_i = h_1(Z_i) - \theta$  for  $h_1(z) = \mathbb{E}(h_2(z, Z_j))$  with  $h_2(z, z') = \mathbb{E}(h(z, z', Q_{ij}))$ . Then

$$Y_{ij} = \theta + X_i + X_j + V_{ij}.$$

Averaging over all observations yields

$$\hat{\theta} = \theta + \frac{2}{n} \sum_{i=1}^n X_i + \bar{V}_n$$

for  $\bar{V}_n = 2/n(n-1) \sum_{i=1}^n \sum_{i < j} V_{ij}$ . The proof of the theorem proceeds in two steps. The first step is to show that  $\bar{V}_n = o_p(n^{-1/2})$ . The second step is to show that, as  $n$  diverges,  $1/\sqrt{n} \sum_{i=1}^n X_i$  converges in distribution to a normal random variable with mean zero and variance  $\sigma^2 > 0$ . From this

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{L} N(0, 4\sigma^2)$$

follows, thereby completing the proof of the theorem.

**Remainder term.** To show that  $\bar{V}_n = o_p(n^{-1/2})$  it is convenient to introduce the variables

$$W_{ij} = Y_{ij} - h_2(Z_i, Z_j), \quad U_{ij} = h_2(Z_i, Z_j) - h_1(Z_i) - h_1(Z_j) + \theta.$$

Then  $\bar{V}_n = \bar{W}_n + \bar{U}_n$  for  $\bar{W}_n = 2/n(n-1) \sum_{i=1}^n \sum_{i < j} W_{ij}$  and  $\bar{U}_n = 2/n(n-1) \sum_{i=1}^n \sum_{i < j} U_{ij}$ . We handle each of these two terms in turn.

Conditional on  $Z_1, \dots, Z_n$ , the variables  $W_{ij}$  are independent across dyads. They have (conditional) mean zero and variance  $\mathbb{E}(W_{ij}^2 | Z_i, Z_j)$ . We therefore see that  $\mathbb{E}(\bar{W}_n) = 0$  and

$$\mathbb{E}(\bar{W}_n^2) = 4 \mathbb{E} \left( \text{var} \left( \frac{\sum_{i=1}^n \sum_{i < j} W_{ij}}{n(n-1)} \middle| Z_1, \dots, Z_n \right) \right) = 4 \frac{\sum_{i=1}^n \sum_{i < j} \mathbb{E}(\mathbb{E}(W_{ij}^2 | Z_i, Z_j))}{n^2(n-1)^2},$$

where we have used the law of total variance in the first step. Notice that, because we have that

$$\mathbb{E}(W_{ij}^2 | Z_i, Z_j) = \mathbb{E}((Y_{ij} - \theta)^2 | Z_i, Z_j) - (h_2(Z_i, Z_j) - \theta)^2 \leq \mathbb{E}((Y_{ij} - \theta)^2 | Z_i, Z_j),$$

it holds that  $\mathbb{E}(\mathbb{E}(W_{ij}^2 | Z_i, Z_j)) = \mathbb{E}(W_{ij}^2) \leq \mathbb{E}((Y_{ij} - \theta)^2)$ , which is uniformly bounded by Assumption 1. Therefore,

$$\mathbb{E}(\bar{W}_n^2) \lesssim \frac{2}{n(n-1)}$$

where, here and later, we use the notation  $A \lesssim B$  to indicate that  $A \leq cB$  for some finite constant  $c > 0$ . Consequently,

$$\bar{W}_n = O_P(n^{-1})$$

follows by Markov's inequality.

Moving on, observe that  $\bar{U}_n$  is a degenerate U-statistic in the variables  $Z_1, \dots, Z_n$ . Hence,  $\mathbb{E}(\bar{U}_n) = 0$ . We now proceed by deriving an upper bound on its variance, that is, on

$$\mathbb{E}(\bar{U}_n^2) = \frac{4}{n^2(n-1)^2} \sum_{i_1=1}^n \sum_{i_1 < j_1} \sum_{i_2=1}^n \sum_{i_2 < j_2} \mathbb{E}(U_{i_1 j_1} U_{i_2 j_2}).$$

Under the conditions stated in Assumption 1, and given the rate of decay of the mixing coefficients  $\alpha_\tau$  being at most of order  $\tau^{-\rho}$ , we have, letting  $\varepsilon = \rho^{2\delta}/(3\delta+5)$ , the upper bound

$$|\mathbb{E}(U_{i_1 j_1} U_{i_2 j_2})| \lesssim \alpha_{\max\{j_1 - i_1, j_2 - i_2\}}^{2\delta/(3\delta+5)} \lesssim \max\{j_1 - i_1, j_2 - i_2\}^{-\varepsilon}$$

by an application of Lemma 3.3 in [Dehling and Wendler \(2010\)](#). Introduce the shorthands

$$m_1 = j_1 - i_1 \geq 1, \quad m_2 = j_2 - i_2 \geq 1.$$

For fixed  $m_1$  there are  $n - m_1$  choices of  $i_1$ , and for fixed  $m_2$  there are  $n - m_2$  choices of  $i_2$ . Thus,

$$\mathbb{E}(\bar{U}_n^2) \lesssim \frac{1}{n^2(n-1)^2} \sum_{m_1=1}^{n-1} \sum_{m_2=1}^{n-1} (n - m_1)(n - m_2) \max\{m_1, m_2\}^{-\varepsilon}.$$

By symmetry of the summand in the pair  $(m_1, m_2)$  we are free to presume that  $m_1 \geq m_2$  and multiply through by two to obtain

$$\mathbb{E}(\bar{U}_n^2) \lesssim \frac{1}{n^2(n-1)^2} \sum_{m_1=1}^{n-1} \sum_{m_2=1}^{m_1} (n - m_1)(n - m_2) m_1^{-\varepsilon}.$$

Note that  $n - m_1 \leq n$  and  $n - m_2 \leq n$ , and there are at most  $m_1$  terms in the inner sum. Hence,  $(n - m_1) \sum_{m_2=1}^{m_1} (n - m_2) \leq n^2 m_1$  and

$$\mathbb{E}(\bar{U}_n^2) \lesssim \frac{1}{(n-1)^2} \sum_{m_1=1}^{n-1} m_1^{1-\varepsilon} \lesssim n^{-\varepsilon},$$

using that  $\sum_{m_1=1}^{n-1} m_1^{1-\varepsilon} \lesssim n^{2-\varepsilon}$ . Observing that  $\varepsilon > 1$  we thus have  $\mathbb{E}(\bar{U}_n^2) = o(n^{-1})$  and so

$$\bar{U}_n = o_P(n^{-1/2})$$

follows by another application of Markov's inequality. We have, therefore, shown that  $\bar{V}_n = o_P(n^{-1/2})$ .

**Leading term.** Having established that

$$\sqrt{n}(\hat{\theta} - \theta) = \frac{2}{\sqrt{n}} \sum_{i=1}^n X_i + o_P(1)$$

we set out to show asymptotic normality. In light of Assumption 1, the sequence  $\{X_i\}$  is zero mean, stationary, and strong mixing. Moreover, we have that  $\mathbb{E}(|X_i|^{2+\delta}) < \infty$  and that the mixing coefficients are of size  $-\rho < -(2+\delta)/\delta$ . These conclusions on the sequence  $\{X_i\}$  follow from an application of standard results (see, e.g., [White 2001](#), Theorem 3.49). Therefore, provided that the long-run variance of the process  $\{X_i\}$  is positive, that is, that, as assumed in the theorem,

$$\sigma^2 = \mathbb{E}(X_i^2) + 2 \sum_{\tau=1}^{\infty} \mathbb{E}(X_i X_{i+\tau}) > 0,$$

we readily obtain that  $1/\sqrt{n} \sum_{i=1}^n X_i \xrightarrow{L} N(0, \sigma^2)$  (see, e.g., [White 2001](#), Theorem 5.20), thereby completing the proof.  $\square$

## Proof of Theorem 2

Throughout the proof we write  $m$  for  $m_{\text{HAC}}$ . We first introduce the infeasible estimator  $\check{\sigma}^2 = \check{\omega}_0 + 2 \sum_{\tau=1}^{m-1} (1 - \tau/m) \check{\omega}_\tau$  that uses the covariance estimators

$$\check{\omega}_\tau = \frac{1}{n - \tau} \sum_{i=1}^{n-\tau} X_i X_{i+\tau}.$$

By virtue of Assumption 1 and the requirement that  $m/n \rightarrow 0$  as  $n \rightarrow \infty$ , it follows from [de Jong and Davidson \(2000, Theorem 2.1\)](#) that  $\check{\sigma}^2 \xrightarrow{P} \sigma^2$  as  $n \rightarrow \infty$ . We then need to show that  $|\check{\sigma}^2 - \hat{\sigma}^2| \xrightarrow{P} 0$  to complete the proof.

To do so we establish that (i)  $|\hat{\omega}_0 - \check{\omega}_0| = o_P(1)$  and that (ii)  $\sum_{\tau=1}^{m-1} |\hat{\omega}_\tau - \check{\omega}_\tau| = o_P(1)$ , which suffices in light of the fact that  $0 \leq (1 - \tau/m) \leq 1$  for all relevant  $\tau$ . To proceed, we begin by recalling, from the proof of Theorem 1, the decomposition of the observations as

$$Y_{ij} = \theta + X_i + X_j + V_{ij}.$$

Then, by averaging and recentering, we obtain  $(\tilde{Y}_i - \bar{Y}_n) = (X_i - \bar{X}_n) + (\tilde{X}_i - \bar{X}_n) + (\tilde{V}_i - \bar{V}_n)$ , for

$$\tilde{X}_i = \frac{1}{n-1} \sum_{j \neq i} X_j, \quad \tilde{V}_i = \frac{1}{n-1} \sum_{j \neq i} V_{ij},$$

and  $\bar{X}_n = 1/n \sum_{i=1}^n \tilde{X}_i$  and  $\bar{V}_n = 1/n \sum_{i=1}^n \tilde{V}_i$ . The sampling error in  $(\bar{Y}_i - \bar{Y}_n)$  as an estimator of  $X_i$  thus equals

$$A_i = (\tilde{Y}_i - \bar{Y}_n) - X_i = -\bar{X}_n + (\tilde{X}_i - \bar{X}_n) + (\tilde{V}_i - \bar{V}_n).$$

For any given  $\tau$ , by an application of the Cauchy-Schwarz inequality and using stationarity,

$$\mathbb{E}(|\hat{\omega}_\tau - \check{\omega}_\tau|) \leq \mathbb{E}(|X_i A_{i+\tau}|) + \mathbb{E}(|X_{i+\tau} A_i|) + \mathbb{E}(|A_i A_{i+\tau}|) \leq 2\sqrt{\mathbb{E}(X_i^2) \mathbb{E}(A_i^2)} + \mathbb{E}(A_i^2).$$

By assumption,  $\mathbb{E}(X_i^2) = O(1)$ . Below we show that  $\mathbb{E}(A_i^2) = O(n^{-1})$ . Then, by Markov's inequality, for any  $\epsilon > 0$ ,

$$\mathbb{P}(|(\hat{\omega}_0 - \check{\omega}_0)| > \epsilon) \leq \frac{\mathbb{E}(|(\hat{\omega}_0 - \check{\omega}_0)|)}{\epsilon} = O(n^{-1/2}),$$

which establishes (i), and, similarly,

$$\mathbb{P} \left( \sum_{\tau=1}^{m-1} |(\hat{\omega}_\tau - \tilde{\omega}_\tau)| > \epsilon \right) \leq \frac{\sum_{\tau=1}^{m-1} \mathbb{E}(|(\hat{\omega}_\tau - \tilde{\omega}_\tau)|)}{\epsilon} = O(mn^{-1/2}),$$

from which (ii) follows given that  $m^2 n^{-1} \rightarrow 0$  as  $n \rightarrow \infty$ .

To bound  $\mathbb{E}(A_i^2)$  it suffices to obtain bounds on the second moment of each of the components of  $A_i$ . First, as,

$$n^2 \mathbb{E}(\bar{X}_n^2) = \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}(X_i X_j) = n\omega_0 + 2 \sum_{\tau=1}^{n-1} (n - \tau) \omega_\tau$$

and  $|\omega_\tau| \lesssim \alpha_\tau^{1/2 \delta/2+\delta} = O(\tau^{-\rho/2 \delta/2+\delta})$  by Corollary 6.17 in [White \(2001\)](#) and  $\alpha_\tau = O(\tau^{-\rho})$ , it follows that

$$\mathbb{E}(\bar{X}_n^2) \lesssim n^{-2} \sum_{\tau=1}^{n-1} (n - \tau) \tau^{-\rho/2 \delta/2+\delta} = O(n^{-1}),$$

on noting that  $2\rho > 3\delta+5/\delta > 2+\delta/\delta$  by Assumption [1](#).

Moving on to  $\mathbb{E}(\tilde{X}_i^2)$  we first expand the sum to write

$$(n-1)^2 \mathbb{E}(\tilde{X}_i^2) = \sum_{j=1}^n \sum_{k=1}^n \mathbb{E}(X_j X_k) - \sum_{k \neq i} \mathbb{E}(X_i X_k) - \mathbb{E}(X_i^2).$$

We note that the first term is equal to  $n^2 \mathbb{E}(\bar{X}_n^2)$ , which has already been shown to be  $O(n)$ , while the third term is  $\mathbb{E}(X_i^2) = \omega_0 = O(1)$ . The middle term, finally, behaves like  $\sum_{\tau=1}^{n-1} \tau^{-\rho/2 \delta/2+\delta}$ , which remains bounded as  $n$  grows large given our rate condition on the mixing coefficients. Therefore,  $\mathbb{E}(\bar{X}_n^2) = O(n^{-1})$  holds.

Next we note that  $\mathbb{E}(\bar{V}_n^2) = O(n^{-1})$  was already shown to hold in the proof of Theorem [1](#). The argument to show that  $\mathbb{E}(\tilde{V}_i^2) = O(n^{-1})$  proceeds in the same way. Moreover, recalling the decomposition  $V_{ij} = U_{ij} + W_{ij}$  from the proof of Theorem [1](#) we can write that

$$\tilde{V}_i = \frac{1}{n-1} \sum_{j \neq i} (U_{ij} + W_{ij}) = \tilde{U}_i + \tilde{W}_i.$$

First, because the  $W_{ij}$  are serially uncorrelated across dyads, it is immediate that we have

$$\mathbb{E}(\tilde{W}_i^2) = \frac{1}{(n-1)^2} \sum_{j \neq i} \sum_{k \neq i} \mathbb{E}(W_{ij} W_{ik}) = \frac{1}{(n-1)^2} \sum_{j \neq i} \mathbb{E}(W_{ij}^2) = O(n^{-1}).$$



Next, for the remaining part, we can again rely on Lemma 3.3 in [Dehling and Wendler \(2010\)](#) to arrive at

$$(n-1)^2 \mathbb{E}(\tilde{U}_i^2) = \sum_{j \neq i} \sum_{k \neq i} \mathbb{E}(U_{ij} U_{ik}) \lesssim \sum_{m_1=1}^{n-1} \sum_{m_2=1}^{n-1} \max\{m_1, m_2\}^{-\epsilon} = 2 \sum_{m_1=1}^{n-1} m_1^{1-\epsilon} = O(n^{2-\epsilon}),$$

for some  $\epsilon > 1$ , in light of our rate requirement on the mixing coefficients. Therefore,  $\mathbb{E}(\tilde{U}_i^2) = O(n^{-\epsilon}) = o(n^{-1})$ . Taken together, the final two rates imply that  $\mathbb{E}(\tilde{V}_i^2) = O(n^{-1})$ .

We have thus shown that  $\mathbb{E}(A_i^2) = O(n^{-1})$ . Then (i) and (ii) both follow and the proof is complete.  $\square$

## Proof of Equation (3.2)

Throughout the proof we write  $m$  for  $m_{\text{CB}}$ . We will derive the bootstrap expectation in the slightly more general setting where we set

$$\dot{Y}_{ij}^* = \dot{Y}_{\phi_i \phi_j} = \begin{cases} Y_{\phi_i \phi_j} & \text{if } \phi_i \neq \phi_j \\ D_{\phi_i} & \text{if } \phi_i = \phi_j \end{cases}$$

for chosen  $D_1, \dots, D_n$  which are non-random conditional on the data. We wish to compute  $\mathbb{E}^*(\bar{Y}_n^*) = 1/n(n-1) \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}^*(\dot{Y}_{\phi_i \phi_j})$ . It will be useful to make the bootstrap expectation explicit. We have

$$\mathbb{E}^*(\bar{Y}_n^*) = \frac{1}{n(n-1)} \sum_{u=1}^b \sum_{i=0}^{m-1} \sum_{v=1}^b \sum_{j=0}^{m-1} \mathbb{E}^*(\dot{Y}_{(\varpi_u+i) \bmod n (\varpi_v+j) \bmod n}) \{i \neq j \text{ if } u = v\},$$

where the indicator function prevents (undefined) entries  $\dot{Y}_{ii}^*$  from arising in the summand. We next split up the right-hand sided into two terms; the first concerns observations within the same block, i.e.,  $u = v$ , the second observations across blocks, i.e.,  $u \neq v$ .

The within-block contribution is

$$\frac{1}{n(n-1)} \sum_{u=1}^b \sum_{i=0}^{m-1} \sum_{j \neq i} \mathbb{E}^*(\dot{Y}_{(\varpi_u+i) \bmod n (\varpi_u+j) \bmod n}).$$

Notice that this term is exactly equal to zero when  $m = 1$ . Thus take  $m > 1$ . As  $\varpi_u$  is uniformly distributed on  $\{1, 2, \dots, n\}$ , this is equal to

$$\frac{b}{n(n-1)} \sum_{i=0}^{m-1} \sum_{j \neq i} \frac{\sum_{k=1}^n \dot{Y}_{(k+i) \bmod n (k+j) \bmod n}}{n}.$$

Choose a pair of indices  $(i, j)$  with  $j - i = d > 0$ . By the circularity of the bootstrap design,

$$\frac{\sum_{k=1}^n \dot{Y}_{(k+i) \bmod n (k+j) \bmod n}}{n} = \frac{\sum_{k=1}^n \dot{Y}_{(k) \bmod n (k+d) \bmod n}}{n} = \frac{\sum_{i=1}^{n-d} Y_{i(i+d)} + \sum_{i=1}^d Y_{(n-d+i)i}}{n},$$

which is the average of the (extended)  $d$ th off-diagonal of the data matrix, say  $\bar{Y}_{(d)}$ . This only depends on  $d$ . Hence, exploiting symmetry of the data, the within-block contribution to  $\mathbb{E}^*(\bar{Y}_n^*)$  equals

$$2 \frac{b}{n(n-1)} \sum_{d=1}^{m-1} (m-d) \bar{Y}_{(d)} = O_P((m-1)(n-1)^{-1});$$

here, the rate holds because  $\bar{Y}_{(d)} = O_P(1)$  uniformly in  $d$  and  $\sum_{d=1}^{m-1} (m-d) = m(m-1)/2$ , and we have used that  $bm = n$ .

The between-block contribution is

$$\frac{1}{n(n-1)} \sum_{u=1}^b \sum_{i=0}^{m-1} \sum_{v \neq u}^{m-1} \sum_{j=0}^{m-1} \mathbb{E}^*(\dot{Y}_{(\varpi_u+i) \bmod n (\varpi_v+j) \bmod n}),$$

observing that the indicator function will always take the value one for observations across different blocks. Here,  $\varpi_u$  and  $\varpi_v$  are independent draws from the uniform distribution on  $\{1, 2, \dots, n\}$ , and so

$$\mathbb{E}^*(\dot{Y}_{(\varpi_u+i) \bmod n (\varpi_v+j) \bmod n}) = \frac{\sum_{k_1=1}^n \sum_{k_2=1}^n \dot{Y}_{(k_1+i) \bmod n (k_2+j) \bmod n}}{n^2} = \frac{\sum_{k_1=1}^n \sum_{k_2=1}^n \dot{Y}_{k_1 k_2}}{n^2}$$

for each pair of blocks  $u \neq v$ . The independence of this expectation on the pair  $(i, j)$  again follows from the circularity of the bootstrap scheme. The between-block contribution, therefore, is

$$\frac{b(b-1)m^2}{n(n-1)} \frac{\sum_{k_1=1}^n \sum_{k_2=1}^n \dot{Y}_{k_1 k_2}}{n^2} = \frac{n-m}{n-1} \left( \frac{n-1}{n} \bar{Y}_n + \frac{1}{n} \bar{D}_n \right) = \bar{Y}_n + O_P(mn^{-1})$$

as long as  $\bar{D}_n = 1/n \sum_{i=1}^n D_i = O_P(1)$ . In the case that  $D_i = 0$  for all  $1 \leq i \leq n$  we arrive at

$$\mathbb{E}^*(\bar{Y}_n^*) = \left(1 - \frac{m}{n}\right) \bar{Y}_n + O_P\left(\frac{m-1}{n}\right),$$

which is Equation (3.2). □

## Proof of Theorems 3 and 4

Throughout the proof we write  $m$  for  $m_{\text{CB}}$ . As in the derivation of Equation (3.2) we will work under the more general setting where we set

$$\dot{Y}_{ij}^* = \dot{Y}_{\phi_i \phi_j} = \begin{cases} Y_{\phi_i \phi_j} & \text{if } \phi_i \neq \phi_j \\ D_{\phi_i} & \text{if } \phi_i = \phi_j \end{cases}$$

for  $D_1, \dots, D_n$  that are non-random conditional on the data. We will only require that  $\mathbb{E}(D_i^2) < \infty$ . As in the proof of Theorem 1 we begin by setting up a Hoeffding-type decomposition for the bootstrap data. This can be done by defining

$$\dot{W}_{ij}^* = \dot{W}_{\phi_i \phi_j} = \begin{cases} Y_{\phi_i \phi_j} - h_2(Z_{\phi_i}, Z_{\phi_j}) & \text{if } \phi_i \neq \phi_j \\ D_{\phi_i} - h_2(Z_{\phi_i}, Z_{\phi_i}) & \text{if } \phi_i = \phi_j \end{cases},$$

and  $\dot{U}_{ij}^* = \dot{U}_{\phi_i \phi_j} = h_2(Z_{\phi_i}, Z_{\phi_j}) - h_1(Z_{\phi_i}) - h_1(Z_{\phi_j}) + \theta$ . Here, the  $D_i$  only appear in the construction of the artificial ‘diagonal’ variables  $\dot{W}_{ii}^*$ . Let  $\bar{W}_n^*$  and  $\bar{U}_n^*$  be the corresponding averages over all observations. Then,

$$\bar{Y}_n^* = \theta + \frac{2}{n} \sum_{i=1}^n X_i^* + \bar{U}_n^* + \bar{W}_n^*.$$

We next recenter this expression by using the decomposition for  $\bar{Y}_n$  from before and multiply through by  $\sqrt{n}$  to arrive at

$$\sqrt{n}(\bar{Y}_n^* - \bar{Y}_n) = \frac{2}{\sqrt{n}} \sum_{i=1}^n (X_i^* - X_i) + \sqrt{n}(\bar{U}_n^* - \bar{U}_n) + \sqrt{n}(\bar{W}_n^* - \bar{W}_n),$$

which is the starting point of our analysis.

**Remainder term.** We first set out to show that

$$\sqrt{n}(\bar{U}_n^* - \bar{U}_n) = o_{P^*}(1), \quad \sqrt{n}(\bar{W}_n^* - \bar{W}_n) = o_{P^*}(1),$$

where, for a random variable  $A_n^*$ ,  $A_n^* = o_{P^*}(1)$  means that  $\mathbb{P}(\mathbb{P}^*(|A_n^*| > \epsilon^*) > \epsilon) = o(1)$  for all  $\epsilon^* > 0$  and  $\epsilon > 0$ . It was already established in the proof of Theorem 1 that  $\bar{U}_n = o_P(n^{-1/2})$  and that  $\bar{W}_n = o_P(n^{-1/2})$ . Further, by Lemma 3.7 in [Dehling and Wendler](#)

(2010) we have that  $n \mathbb{E}(\mathbb{E}^*(\bar{U}_n^{*2})) = o(1)$ . From this  $\bar{U}_n^* = o_{P^*}(n^{-1/2})$  follows by repeated application of Markov's inequality. It then only remains to verify that  $\bar{W}_n^* = o_{P^*}(n^{-1/2})$ . We turn to doing so next.

As in the proof of Equation (3.2), we begin by writing out the sample average in full as

$$\bar{W}_n^* = \frac{1}{n(n-1)} \sum_{u=1}^b \sum_{v=1}^b \sum_{i=0}^{m-1} \sum_{j=0}^{m-1} \dot{W}_{(\varpi_u+i) \bmod n (\varpi_v+j) \bmod n} \{i \neq j \text{ if } u = v\}.$$

Hence,  $\mathbb{E}^*(\bar{W}_n^{*2})$  is the (normalized) sum of

$$\mathbb{E}^*(\dot{W}_{(\varpi_{u_1}+i_1) \bmod n (\varpi_{v_1}+j_1) \bmod n} \dot{W}_{(\varpi_{u_2}+i_2) \bmod n (\varpi_{v_2}+j_2) \bmod n}) \begin{cases} i_1 \neq j_1 \text{ if } u_1 = v_1 \\ i_2 \neq j_2 \text{ if } u_2 = v_2 \end{cases}$$

over all  $b^4$  combinations of blocks  $u_1, u_2, v_1, v_2$  and all  $m^4$  combinations of indices  $i_1, j_1, i_2, j_2$ . The form of this bootstrap expectation depends on whether some of the indices coincide and, if so, which ones.

First consider the situation where all four blocks  $u_1, v_1, u_2, v_2$  are distinct. Then the  $\varpi_{u_1}, \varpi_{v_1}, \varpi_{u_2}, \varpi_{v_2}$  are all independent uniform random variables on  $\{1, 2, \dots, n\}$  and so (again exploiting cyclicity of the bootstrap scheme) the bootstrap expectation is equal to

$$\left( \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \dot{W}_{ij} \right)^2 = \left( \frac{n-1}{n} \bar{W}_n + \frac{1}{n} \bar{H}_n \right)^2 = \frac{(n-1)^2}{n^2} \bar{W}_n^2 + 2 \frac{(n-1)}{n^2} \bar{W}_n \bar{D}_n + \frac{1}{n^2} \bar{D}_n^2,$$

where (relative to the proof of Equation (3.2)) we redefine  $\bar{D}_n = 1/n \sum_{i=1}^n D_i - h_2(Z_i, Z_i)$ . This does not depend on any of the indices under consideration. Furthermore, from the proof of Theorem 1 we know that  $\mathbb{E}(\bar{W}_n^2) = O(n^{-2})$ . Provided that  $\mathbb{E}(D_i^2) < \infty$  we also have that  $\mathbb{E}(\bar{D}_n^2) = O(1)$  because  $\mathbb{E}(h_2(Z_i, Z_i)^2) < \infty$  by Assumption 1. We thus have that

$$\mathbb{E} \left( \left( \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \dot{W}_{ij} \right)^2 \right) = O(n^{-2}).$$

There are  $b!/(b-4)!$  combinations of four distinct blocks and  $m^4$  different combinations of indices  $i_1, j_1, i_2, j_2$ . The total contribution of distinct-block terms to  $\mathbb{E}(\mathbb{E}^*(\bar{W}_n^{*2}))$  is then at most of order  $n^{-2}$ .

Next consider the opposite extreme where all observations are from the same block, i.e.,  $u_1 = v_1 = u_2 = v_2$ . For such terms to contribute we need that  $i_1 \neq j_1$  and  $i_2 \neq j_2$ . Now the bootstrap expectation of a representative term of this configuration has as expectation

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E}(W_{(k+i_1) \bmod n (k+j_1) \bmod n} W_{(k+i_2) \bmod n (k+j_2) \bmod n})$$

which is non-zero only when  $i_1 = i_2$  and  $j_1 = j_2$ , in which case it equals  $\mathbb{E}(W_{i_1 j_1}^2) < \infty$ . Therefore, for this second case, a contribution to  $\mathbb{E}(\mathbb{E}^*(\bar{W}_n^{*2}))$  comes only from  $bm(m-1)$  terms. Normalized by  $n^2(n-1)^2$  this then gives an order of magnitude of  $mn^{-3} = o(n^{-2})$  given that  $mn^{-1} \rightarrow 0$ .

Then there are a variety of different configurations where some but not all observations belong to the same block. These configurations feature either (i) two pair of observations with pairs in different blocks but observations within a pair in the same block, or (ii) a triplet of observations in the same block and the fourth observation in a different block. The different cases for each configuration have the same structure and so can be handled in the same way.

For the first of these two configurations, consider the case where  $u_1 = v_1$  and  $u_2 = v_2$ , but  $u_1 \neq v_2$ . There are  $b(b-1)$  such group pairings. A representative configuration of  $i_1, j_1, i_2, j_2$  has  $i_1 \neq j_1$  and  $i_2 \neq j_2$  and, therefore, the bootstrap expectation here is equal to

$$\frac{1}{n^2} \sum_{k_1=1}^n \sum_{k_2=1}^n W_{(k_1+i_1) \bmod n (k_1+j_1) \bmod n} W_{(k_2+i_2) \bmod n (k_2+j_2) \bmod n}.$$

The summand here has zero expectation unless (i)  $k_1 + i_1 = k_2 + i_2$  and  $k_1 + j_1 = k_2 + j_2$  or (by symmetry) (ii)  $k_1 + i_1 = k_2 + j_2$  and  $k_1 + j_1 = k_2 + i_2$ . Take (i), which is equivalent to having  $k_1 = k_2 + d$  for each admissible value  $d = i_2 - i_1 = j_2 - j_1$ . The bootstrap expectation equals  $1/n^2 \sum_{k=1}^n W_{k, k+d}^2$ , whose expectation is  $O(n^{-1})$  for any  $d$ . Ranging over all  $d$  yields  $O(m^4)$  index combinations. Hence, the contribution of this case to  $\mathbb{E}(\mathbb{E}^*(\bar{W}_n^{*2}))$  is of the order  $b^2 m^4 n^{-5} = m^2 n^{-3} = o(n^{-1})$ . By symmetry, the same conclusion holds for (ii).

For the second of the configurations we can take  $u_1 = v_1 = u_2 = u$  and  $v_2 = v$  for some  $u \neq v$ . There are again  $b(b-1)$  such permissible group pairs. In this case the indicator

function ensures that we will only face terms with  $i_1 \neq j_1$ . The bootstrap expectation for a given collection of permissible indices  $i_1, j_1, i_2, j_2$  for this type of configuration then equals

$$\frac{1}{n^2} \sum_{k_1=1}^n \sum_{k_2=1}^n W_{(k_1+i_1) \bmod n} W_{(k_1+j_1) \bmod n} \dot{W}_{(k_1+i_2) \bmod n} \dot{W}_{(k_2+j_2) \bmod n}.$$

We distinguish two cases. The first case has  $(k_1 + i_2) \bmod n \neq (k_2 + j_2) \bmod n$ . Then the average does not feature any artificial diagonal entries and the expectation of the summand is zero unless (i)  $i_1 = i_2 = i$  and  $(k_1 + j_1) \bmod n = (k_2 + j_2) \bmod n$  or (ii)  $(k_1 + i_1) \bmod n = (k_2 + j_2) \bmod n$  and  $i_2 = j_2$ . For Case (i), the expectation of the above sum becomes

$$\frac{1}{n^2} \sum_{k_1=1}^n \mathbb{E}(W_{(k_1+i) \bmod n}^2) = O(n^{-1}).$$

Because there are at most  $m^3$  such terms for each of the  $b(b-1)$  block pairs the total contribution of such configuration is of the order  $b^2 m^3 n^{-5} = (mn^{-1})n^{-2} = o(n^{-2})$ . For Case (ii), in turn, we arrive at

$$\frac{1}{n^2} \sum_{k_1=1}^n \mathbb{E}(W_{(k_1+i_1) \bmod n} W_{(k_1+j_1) \bmod n} (D_{(k_1+i_2) \bmod n} - h_2(Z_{(k_1+i_2) \bmod n}, Z_{(k_1+i_2) \bmod n})))$$

which, by Cauchy-Schwarz, is easily seen to be  $O(n^{-1})$ . Here, for each of the  $b(b-1)$  block pairings there are  $m^3$  such terms yielding a total contribution to  $\mathbb{E}(\mathbb{E}^*(\bar{W}_n^{*2}))$  that is again  $o(n^{-2})$ .

Having shown that  $n \mathbb{E}(\mathbb{E}^*(\bar{W}_n^{*2})) = o(1)$  it follows that  $\bar{W}_n^* = o_{P^*}(n^{-1/2})$  by Markov's inequality. This completes our analysis of the remainder term in the bootstrap version of Hoeffding's decomposition.

**Bootstrap consistency.** From the previous step,

$$\sqrt{n}(\bar{Y}_n^* - \bar{Y}_n) = 2\sqrt{n}(\bar{X}_n^* - \bar{X}_n) + o_{P^*}(1),$$

with  $\bar{X}_n^* = 1/n \sum_{i=1}^n X_i^*$ , and it suffices to show that

$$(i) \text{ var}^*(\sqrt{n}\bar{X}_n^*) = n \mathbb{E}^*((\bar{X}_n^* - \bar{X}_n)^2) \xrightarrow{P} \sigma^2,$$

(ii)  $\sup_a |\mathbb{P}^*(\sqrt{n}(\bar{X}_n^* - \bar{X}_n) \leq a) - \mathbb{P}(\sqrt{n}\bar{X}_n \leq a)| \xrightarrow{P} 0$  as  $n \rightarrow \infty$ .

As  $\{X_i\}$  is a stationary mixing process both (i) and (ii) follow readily from the bootstrap literature on time series. Here, we use Theorem 1 [Calhoun \(2018\)](#). This completes the proof of both Theorem [3](#) and Theorem [4](#).  $\square$

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